

Markscheme

Mathematics: analysis and approaches

Practice question bank

31. (a)

Consider the function $f(x) = \frac{3x^2 + 7x - 3}{(x + 2)(x^2 + 2x + 5)}$.

Write $f(x)$ as the sum of partial fractions in the form $\frac{A}{x + 2} + \frac{Bx + C}{x^2 + 2x + 5}$.

attempt to write $A(x^2 + 2x + 5) + (Bx + C)(x + 2)$ (M1)

$$3x^2 + 7x - 3 = A(x^2 + 2x + 5) + (Bx + C)(x + 2) \quad \text{A1}$$

substituting $x = -2$: $-5 = 5A \Rightarrow A = -1$ A1

comparing coefficients of x^2 and constant terms (M1)

$$3 = A + B \Rightarrow B = 4 \quad \text{A1}$$

$$-3 = 5A + 2C \Rightarrow C = 1 \quad \text{A1}$$

[6 marks]

(b)

Hence, by differentiating your partial fraction form term by term, find $f'(0)$.

differentiate each term (quotient rule on the second term) (M1)

$$f'(x) = \frac{1}{(x + 2)^2} + \frac{-4x^2 - 2x + 18}{(x^2 + 2x + 5)^2} \quad \text{A1}$$

$$f'(0) = \frac{1}{4} + \frac{18}{25} = \frac{97}{100} = 0.97 \quad \text{A1}$$

[3 marks]